

CYBER SECURITY

Core Certification Course Outline I Syllabus 2.0

Course Overview: This module covers essential concepts for the secure use of ICT in daily life. Candidates will learn to maintain secure network connections, use the Internet safely, protect against malware and cyber threats, manage passwords effectively, and handle data securely through proper backup and deletion procedures.

Core Competencies Developed

Data Protection	Core Competencies Developed			Network Security	Access Control
	 Malware Defense	 Network Security	 Access Control		
1 Security Concepts • Data vs information and cybercrime fundamentals • Threats: malicious, accidental, natural disasters • Cloud computing security risks and data control • Information security: confidentiality, integrity, availability • Identity theft prevention and social engineering • Data protection principles and GDPR compliance • File encryption and password protection techniques 2 Malware • Understanding malware types and concealment methods • Trojans, rootkits, backdoors, viruses, and worms • Ransomware, spyware, adware, and botnets • Keystroke logging and dialler threats • Anti-virus software functions and limitations • Software updates and patch management • Scanning, quarantining, and removing infections • Risks of obsolete and unsupported software 3 Network Security • Network types: LAN, WLAN, WAN, VPN • Security implications of network connections • Network administrator roles and responsibilities • Firewall functions in personal and work environments • Configuring firewall rules and permissions • Wireless security protocols: WEP, WPA, WPA2 • MAC filtering and SSID hiding techniques • Personal hotspots and secure device connections 4 Access Control • Preventing unauthorised access to data • Multi-factor authentication and one-time passwords • Network account management and security • Biometric security: fingerprint, face, eye scanning • Password policies and best practices • Password length, complexity, and uniqueness • Password manager software benefits and limitations 5 Secure Web Use • Browser settings: autocomplete and autosave controls • Deleting private data: history, cookies, cached files • Secure web pages for banking and online purchases • Website authentication and validation techniques • Security certificates and domain verification • Understanding pharming attacks • Content-control and parental control software 6 Communications • Email encryption and digital signatures • Identifying phishing and fraudulent emails • Reporting phishing to authorities • Malware risks in email attachments • Social networking privacy and security settings • Cyber bullying, grooming, and false identities • VoIP and instant messaging vulnerabilities • Mobile app permissions and data extraction risks • Lost device measures: remote wipe and disable					

7 Secure Data Management

- Physical security: cable locks and access control
- Backup procedures and scheduling strategies
- Backup locations: local, external, cloud storage
- Data restoration from various backup sources
- Deleting vs permanently deleting data
- Permanent deletion from social media and cloud
- Data destruction methods: shredding, degaussing
- Using data destruction utilities effectively

